tRNA Guanine Transglycosylase from *S. aureus* Enables Efficient Site-Specific Covalent Labeling of DNA

Zhenglong Zhai, Alexander Harjung, Evan McCormack and Neal K. Devaraj*[a]

[a] Z. Zhai, A. Harjung, E. McCormack, Prof. N. K. Devaraj
Department of Chemistry and Biochemistry, University of California, San Diego, 9500 Gilman Drive, La Jolla, California 92093, United States
E-mail: ndevaraj@ucsd.edu

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Abstract: Bacterial tRNA guanine transglycosylases (TGTs) exchange guanine-34 in cognate tRNAs for prequeuosine1 (preQ1). *Escherichia coli* TGT has been widely harnesses for site-specific RNA labeling and, through substrate optimization, covalent DNA labeling. However, *E. coli* TGT exhibits slow labeling kinetics and low efficiency on DNA. Here, we demonstrate that TGT from *Staphylococcus aureus* is a superior enzyme for DNA modification. Like *E. coli* TGT, *S. aureus* TGT recognizes only the anticodon stem-loop and efficiently modifies the DNA hairpin dECYMH, whereas *E. coli* TGT shows almost no activity. Using *S. aureus* tRNA-derived sequences as starting points, we identified the $tRNA^{His}$-based hairpin as the most promising scaffold. Rational loop optimization yielded dSAH-6, which is labeled by *S. aureus* TGT with near-quantitative conversion under standard conditions. Side-by-side comparisons revealed that *S. aureus* TGT outperforms *E. coli* TGT in both yield and kinetics across multiple preQ1 derivatives and hairpin sequences. Importantly, the optimized dSAH-6 sequence enables robust labeling at 5'-, 3'-, or internal positions in longer ssDNA constructs and supports efficient multi-site incorporation for enhanced detection sensitivity. These advances significantly expand the DNA-TAG toolbox and highlight how natural variation in TGT substrate promiscuity can be exploited for improved nucleic acid modification technologies.

Introduction

Labeling of nucleic acids is essential for many studies aiming to elucidate structure, function, and dynamics *in vitro* and in living cells[1]. Among the numerous labeling strategies that have been developed, covalent labeling is particular attractive because it provides the most stable linkage, an unrivaled choice of small and highly fluorescent labels and an incredible versatility. Chemoenzymatic methods offer several advantages since it occurs in mild and aqueous conditions and achieves exquisite site-specificity through the natural specificity of enzymes[2,3]. Enhancing enzymatic tools for covalent, site-specific nucleic acid labeling under physiological conditions holds the tremendous promise for applications in RNA and DNA biology, molecular diagnostics and therapeutics[4,5].

Our group has previously developed *Escherichia coli* tRNA guanine transglycosylases (TGT) as a powerful tool for site-specific RNA labeling. *E. coli* TGT catalyzes the exchange of a guanine at the wobble position of the anticodon loop of tRNA with 7-deazaguanine derivatives. Inspired by the mechanism, *E. coli* TGT has been used for incorporating preQ1 substrate which is conjugated to functional groups, such as fluorophores, affinity handles, and biorthogonal groups, onto its 17 or 25 mer RNA hairpin. *E. coli* TGT has been harnesses for numerous biotechnological applications, including light activated control of translation and CRISPR gene editing[6,7], affinity purification and identification of RNA-protein interactions[8], and protein-RNA conjugation[9].

More Recently, we extended this strategy to DNA by optimizing the DNA hairpin substrates to minimize the steric hindrance around TGT active site[10]. However, prior efforts to mutagenize the *E. coli* TGT active site failed to produce an enzyme that could efficiently incorporate preQ1 derivatives onto DNA hairpins. This strong preference of *E. coli* TGT for RNA substrate, as well as the difficulty in engineering *E. coli* TGT to be more permissive for its substrate, eventually led us to a different approach: we decided to explore TGTs from other organisms to see if we could identify an enzyme with increase promiscuity. TGTs are present across all kingdoms of life and exhibit variation in substrate specificity. Bacteria TGTs typically incorporate preQ1 at the wobble position of tRNA anticodon loop, eukaryotic TGTs insert queuine directly, and archaeal TGTs modify a site in the D-arm with the alternative base preQ0. However, these preferences are not rigid, and evolutionary adaptation has produced notable exceptions. Recent studies have shown that, certain bacteria TGTs from intracellular human pathogens, such as *Chlamydia trachomatis*, have shifted their substrate specificity from preQ1 to queuine, while others, such as the facultative intracellular pathogen *Bartonella henselae*, exhibit rare dual-substrate specificity and can accept both preQ1 and queuine[11,12]. Thus, exploring TGTs from diverse species may enable new powerful modification reactions.

Herein, we demonstrate the TGT from *Staphylococcus aureus*, a common Gram-positive bacterium that causes a wide variety of clinical infections, can serve as an alternative enzyme for both RNA transglycosylation at deoxyguanosine (RNA-TAG) and especially DNA transglycosylation at deoxyguanosine (DNA-TAG). Through iterative testing of rationally designed sequences, we achieved high efficiency incorporation of preQ1 probes into 25-mer DNA hairpins. Next, we characterized the ability of DNA-TAG to label DNA constructs containing one or more recognition sequences. Finally, we compared the labeling kinetics of *S. aureus* TGT and *E. coli* TGT toward optimized DNA substrate and found that *S. aureus* performed better. *S. aureus* TGT thus expands our DNA-TAG toolbox, highlights the differential substrate promiscuity of TGTs from different species and open the door to novel DNA and RNA applications.

Results and Discussion

S. aureus TGT can incorporate preQ1 derivatives into tRNA, RNA and DNA hairpin.

In previous research, we demonstrated that *E. coli* TGT can incorporate preQ1 derivative into both RNA and DNA substrates (Figure 1A, B)[10,13]. Labeling is monitored by a gel-shift assay that exploits the molecular-weight increase upon covalent modification of the preQ1 conjugate. Successful enzymatic exchange of guanine for preQ1-derivatives results in a clear upward shift of the product band relative to the unmodified oligonucleotide (Figure 1C). Although *E. coli* TGT accepts DNA as substrate, the labeling efficiency remains suboptimal, and the complete modification typically requires about 4 hours.

S. aureus is a Gram-positive bacterial pathogen responsible for a broad spectrum of clinical infections, ranging from minor infections to life-threatening systemic diseases. Notably, methicillin-resistant *S. aureus* (MRSA) strains are a major cause of healthcare-associated and community-acquired infections worldwide[14]. Although *E. coli* TGT is the most widely used and well-characterized, structure information for it and most of its homologs, including *S. aureus* TGT, remains limited. Only a few crystal structures from other species, such as *Z. mobilis* and human, are available[15,16]. Sequence analysis reveals that *S. aureus* TGT shares 35% sequence identity with *E. coli* one (Figure S1). We used AlphaFold3 to predict their structures and found that *S. aureus* TGT has a larger active site cavity than *E. coli* one (Figure S2). The structure variations raised the possibility that *S. aureus* TGT might exhibit greater substrate tolerance.

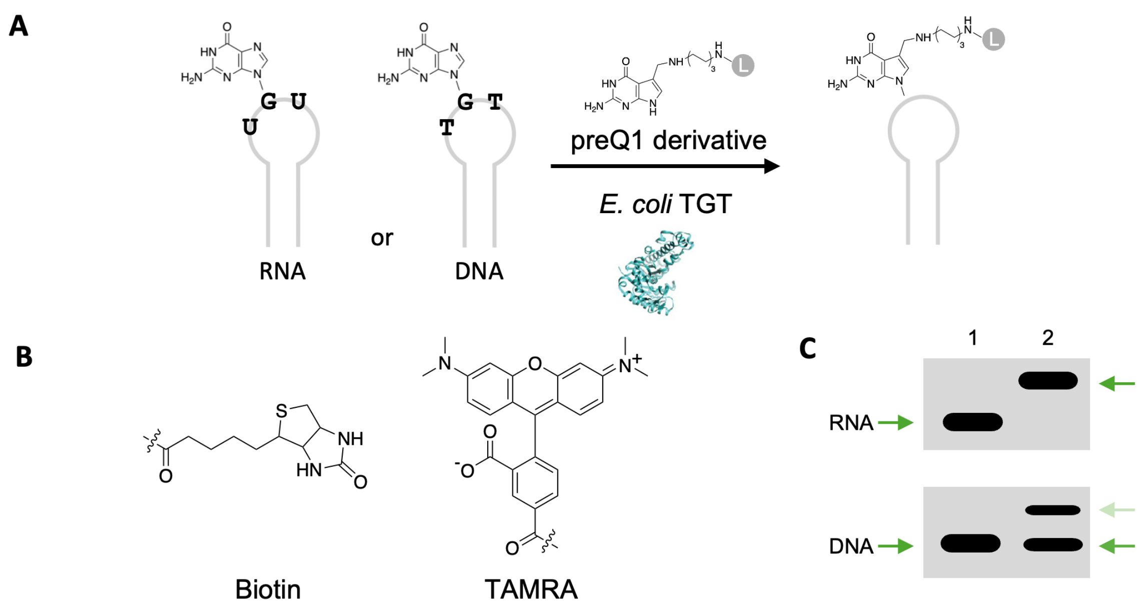

Figure 1. *E. coli* TGT mediated RNA or DNA modification with functional small molecules. (A) General scheme of RNA-TAG and DNA-TAG. *E. coli* TGT irreversibly exchanges a specific guanine with a preQ1-ligand (L) on an RNA or DNA hairpin. (B) PreQ1-ligands used in this work include preQ1-biotin, preQ1-tetramethylrhodamine (TAMRA). (C) Cartoon depiction of expected upward gel shift due to increased mass from the base exchange reaction as indicated by the green arrows. DNA labeling by *E. coli* TGT is less efficient than RNA labeling under the same condition. Lane 1: DNA or RNA start material; Lane 2: preQ1-L modified DNA or RNA.

S. aureus TGT had not been previously expressed or purified. The coding sequence of *S. aureus* TGT was cloned into a bacterial expression plasmid with a N-terminal hexahistidine tag and C-terminal Strep-tag for affinity purification. To enhance soluble expression, an N-terminal maltose-binding protein (MBP) tag was also included for expression and cleaved during purification. We subsequently over-expressed *S. aureus* TGT in *E. coli* and performed dual purification as describe in the Methods section (Figure S3).

Since four tRNAs with GUN anticodons are natural substrates for TGT, including tRNA^{Tyr}, tRNA^{His}, tRNA^{Asp}, and tRNA^{Asn}, we first tested whether *S. aureus* TGT displayed activity toward its cognate full-length tRNA substrates *in vitro*. The result showed that it efficiently labeled both tRNA^{Tyr} and tRNA^{His} with preQ1-TAMRA (Figure 2A). We next tested it on 25-mer hairpins, and it successfully incorporated preQ1-biotin onto extended-stem tRNA^{Tyr}-based RNA mini helix (ECYMH), although with slightly lower efficiency than *E. coli* one. Interestingly, *S. aureus* TGT exhibited markedly superior labeling of the DNA analog dECYMH, whereas *E. coli* TGT showed almost no activity (Figure 2B). From an evolutionary perspective, *S. aureus* TGT may have greater compatibility with RNA or DNA hairpins derived from its own organism's tRNAs. We there reasoned that hairpin substrates derived from these alternative *S. aureus* tRNAs would provide a good start point for developing improved DNA-TAG substrate for *S. aureus* TGT. Following established nomenclature, we designated the 25-nuceotide wide-type hairpins as dSAH-0, dSAY-0, dSAD-0, dSAN-0, corresponding to the hairpins derived from tRNA^{His}, tRNA^{Tyr}, tRNA^{Asp}, and tRNA^{Asn} from *S. aureus*, respectively. Although all four cognate tRNAs share the identical UGU recognition sequence, the flanking anti-codon loop residues differ significantly. All hairpins were treated with preQ1-biotin and TGT. Interestingly, both TGTs showed clear labeling of dSAH-0 but virtually no activity toward the other three substrates (Figure 2C). This preference for tRNA^{His} derived hairpin mirrors our previous finding with *E. coli* TGT and confirms that this hairpin is the most promising scaffold for further optimization[10].

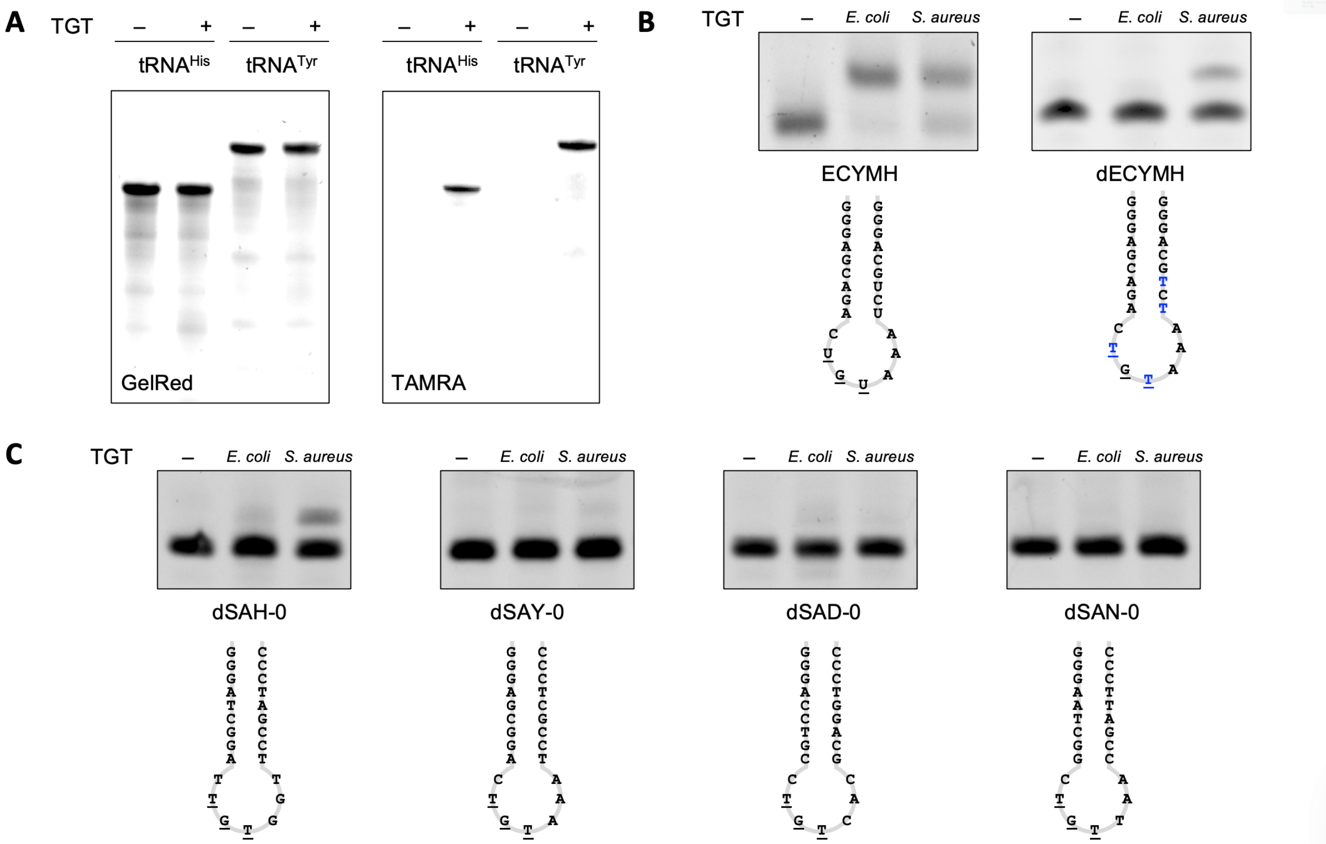

Figure 2. TGT-mediated labeling of tRNA and hairpin substrates. (A) *S. aureus* TGT labeling of tRNA substrates from *S. aureus* using preQ1-tetramethylrhodamine (TAMRA). (B) *E. coli* and *S. aureus* TGT labeling of RNA hairpin ECYMH and its DNA counterpart dECYMH using preQ1-biotin. While both *S. aureus* TGT and *E. coli* TGT are able to modify the ECYMH RNA hairpin, only *S. aureus* TGT showed labeling for the corresponding DNA-hairpin (dECYMH). (C) *E. coli* and *S. aureus* TGT labeling of TAG3 hairpins derived from cognate tRNA substrates from *S. aureus* using preQ1-biotin. *S. aureus* TGT showed clear labeling of tRNA^{His}-derived hairpin dSAH-0.

DNA hairpin substrate optimization for *S. aureus* TGT

Encouraged by the result, we decided to use dSAH-0 as the starting point for further optimization to achieve higher labeling efficiency. Previous studies with *E. coli* TGT demonstrated that low DNA labeling efficiency results from steric hindrance rather than fundamental incompatibility, and that the anti-codon loop sequence is dominant determinant of recognition and efficiency. We therefore focused our mutations on the loop region. Guided by rational DNA substrate optimization for *E. coli* TGT, we introduced loop mutations previously identified as optimal for *E. coli* TGT (Figure 3A). Among the variants tested, dSAH-6 exhibited the highest labeling efficiency (>80%) with preQ1-biotin and was selected for all subsequent experiments (Figure 3B).

For practical DNA-TAG applications, the TGT recognition hairpin must be compatible with diverse positions within longer DNA constructs. We therefore designed 67-mer single-stranded DNA oligonucleotides containing the dSAH-6 sequence at the 5' end, 3' end, or internally. All three constructs were quantitatively labeled with preQ1-biotin by *S. aureus* TGT (Figure 3C), confirming the versatility of the optimized hairpin. To enable applications requiring enhanced sensitivity, we explored multi-site labeling. Constructs bearing two tandem dSAH-6 hairpins were fully labeled by *S. aureus* TGT, whereas a control containing a single active hairpin and one inactivated by a TGT→TCT mutation showed only a single gel shift (Figure 3D). In contrast, *E. coli* TGT typically yields incomplete labeling when the recognition hairpin is placed at the 5' terminus, highlighting the superior positional tolerance of *S. aureus* TGT[10].

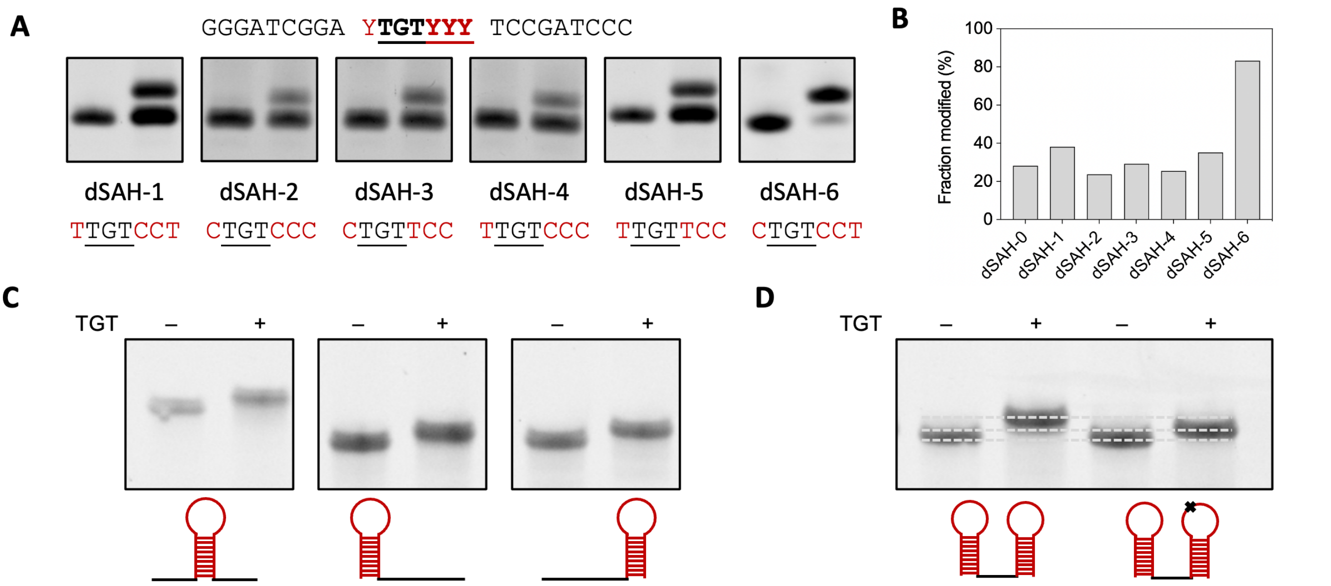

Figure 3. *S. aureus* TGT mediated labeling of DNA. (A) *S. aureus* TGT labeling of mutants. Unmodified oligos are on the left, and reaction products are on the right. Modification is determined by an upward gel shift of the oligo after insertion of preQ1-biotin. (B) Quantification of labeling efficiency for mutants. *S. aureus* TGT modifies dECY-6 at ~80% efficiency. (C) dSAH-6 construct placement. Efficient labeling occurs at either end or internally within ssDNA construct. (D) Insertion of multiple labels. Multiple probes can be incorporated in tandem for enhanced sensitivity, confirmed by increased gel shift as compared to ΔC mutation indicated as X.

Comparison of Labeling efficiency between *S. aureus* TGT and *E. coli* TGT

To directly compare the performance of *S. aureus* TGT and *E. coli* TGT, we conducted side-by-side labeling experiments. Using 25-mer DNA hairpin substrates, *S. aureus* TGT exhibited superior labeling efficiency not only with its own optimized substrate dSAH-6 but also with dECH-10, the sequence previously optimized for *E. coli* TGT (Figure 4A, B). To gain deeper insight into their kinetic differences, we performed time-course experiments with two commonly used preQ1 derivatives. With both preQ1-biotin and preQ1-TAMRA, *S. aureus* TGT displayed significantly faster labeling kinetics and higher final yields than *E. coli* TGT on the respective optimized DNA hairpins. For preQ1-biotin, the half-labeling time ($t_{1/2}$) for reaching saturation was around 45 min for *S. aureus* TGT, whereas *E. coli* TGT showed only relative weak labeling under the same experimental conditions (Figure 4C). For preQ1-TAMRA substrate, *S. aureus* TGT showed around 3.5-fold faster initial rate according to the TAMRA fluorescence signals (Figure S4) and 1.5-fold higher final labeling efficiency than *E. coli TGT* (Figure 4D). These results establish *S. aureus* TGT as a markedly improved enzyme for DNA-TAG applications.

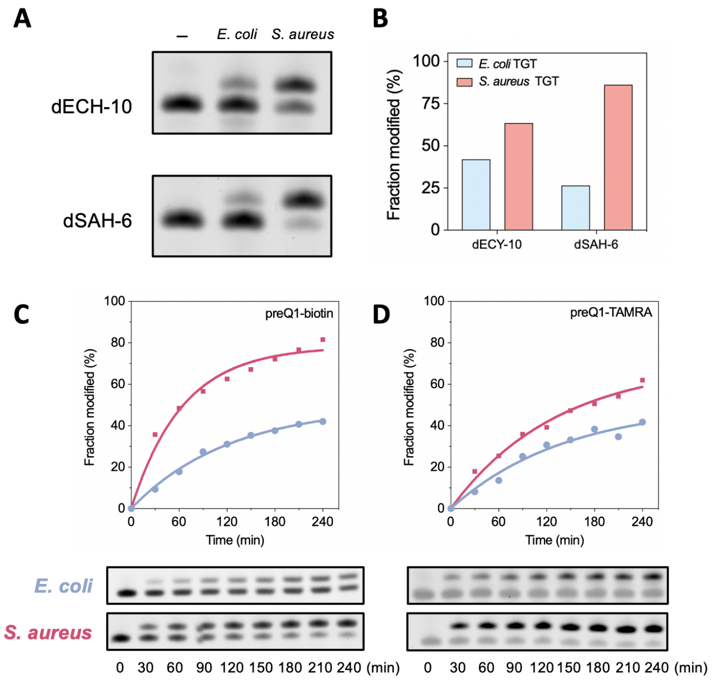

Figure 4. Comparison of *E. coli* and *S. aureus* TGT for DNA labeling. (A) Labeling of dECH-10 and dSAH-6 hairpins with preQ1-biotin by *E. coli* and *S. aureus* TGT. (B) Quantification of labeling efficiency from (A). (C) Labeling kinetics of preQ1-biotin; (D) Labeling kinetics of preQ1-TMR. *S. aureus* TGT exhibits significantly faster kinetics and higher final yields than *E. coli* TGT on both optimized DNA substrates.

Conclusion

In this study, we report the first expression, purification, and application of TGT from *S. aureus* and demonstrate its superior performance for site-specific covalent DNA labeling. We discovered that *S. aureus* TGT exhibits markedly higher tolerance for DNA hairpin substrates than the widely used *E. coli* TGT. These findings illustrate how natural variation in TGT substrate promiscuity can be directly harnessed to overcome limitations of existing enzymes. *S. aureus* TGT significantly expands the DNA-TAG toolbox, transforming a previously inefficient reaction into a practical and versatile technology. The new enzyme opens exciting opportunities for nucleic acid diagnostics, therapeutics, imaging, nanotechnology, and chemical barcoding applications. Further screening of TGTs from diverse species may yield additional variants with unique properties, further enriching enzymatic nucleic acid modification strategies.

Supporting Information

The authors have cited additional references within the Supporting Information[10,17].

Keywords: DNA labeling • DNA • tRNA Guanine Transglycosylase • *S. aureus* • *E. coli*

[1] N. Klöcker, F. P. Weissenboeck, A. Rentmeister, "Covalent labeling of nucleic acids" *Chem Soc Rev* **2020**, *49*, 8749–8773.
[2] D. Gillingham, R. Shahid, "Catalysts for RNA and DNA modification" *Curr Opin Chem Biol* **2015**, *25*, 110–114.
[3] J. M. Holstein, A. Rentmeister, "Current covalent modification methods for detecting RNA in fixed and living cells" *Methods* **2016**, *98*, 18–25.
[4] P. Le, N. Ahmed, G. W. Yeo, "Illuminating RNA biology through imaging" *Nat Cell Biol* **2022**, *24*, 815–824.
[5] L. Li, S. Xu, H. Yan, X. Li, H. S. Yazd, X. Li, T. Huang, C. Cui, J. Jiang, W. Tan, "Nucleic Acid Aptamers for Molecular Diagnostics and Therapeutics: Advances and Perspectives" *Angew Chem Int Ed Engl* **2021**, *60*, 2221–2231.
[6] D. Zhang, L. Liu, S. Jin, E. Tota, Z. Li, X. Piao, X. Zhang, X.-D. Fu, N. K. Devaraj, "Site-Specific and Enzymatic Cross-Linking of sgRNA Enables Wavelength-Selectable Photoactivated Control of CRISPR Gene Editing" *J. Am. Chem. Soc.* **2022**, *144*, 4487–4495.
[7] D. Zhang, C. Y. Zhou, K. N. Busby, S. C. Alexander, N. K. Devaraj, "Light-Activated Control of Translation by Enzymatic Covalent mRNA Labeling" *Angew Chem Int Ed* **2018**, *57*, 2822–2826.
[8] K. N. Busby, A. Fulzele, D. Zhang, E. J. Bennett, N. K. Devaraj, "Enzymatic RNA Biotinylation for Affinity Purification and Identification of RNA–Protein Interactions" *ACS Chem. Biol.* **2020**, *15*, 2247–2258.
[9] E. M. Tota, N. K. Devaraj, "RNA-TAG Mediated Protein-RNA Conjugation" *ChemBioChem* **2023**, *24*, e202300454.
[10] E. M. Tota, N. K. Devaraj, "Site-Specific Covalent Labeling of DNA Substrates by an RNA Transglycosylase" *J. Am. Chem. Soc.* **2023**, *145*, 8099–8106.
[11] Y. Yuan, R. Zallot, T. L. Grove, D. J. Payan, I. Martin-Verstraete, S. Šepić, S. Balamkundu, R. Neelakandan, V. K. Gadi, C.-F. Liu, M. A. Swairjo, P. C. Dedon, S. C. Almo, J. A. Gerlt, V. de Crécy-Lagard, "Discovery of novel bacterial queuine salvage enzymes and pathways in human pathogens" *Proceedings of the National Academy of Sciences* **2019**, *116*, 19126–19135.
[12] S. Quaiyum, Y. Yuan, G. Sun, R. M. M. N. Ratnayake, G. Hutinet, P. C. Dedon, M. F. Minnick, V. de Crécy-Lagard, "Queuosine salvage in Bartonella henselae Houston 1: a unique evolutionary path" *Microbiology* **2024**, *170*, 001490.
[13] S. C. Alexander, K. N. Busby, C. M. Cole, C. Y. Zhou, N. K. Devaraj, "Site-Specific Covalent Labeling of RNA by Enzymatic Transglycosylation" *J. Am. Chem. Soc.* **2015**, *137*, 12756–12759.
[14] R. Touaitia, A. Mairi, N. A. Ibrahim, N. S. Basher, T. Idres, A. Touati, "Staphylococcus aureus: A Review of the Pathogenesis and Virulence Mechanisms" *Antibiotics* **2025**, *14*, 470.
[15] R. Brenk, M. T. Stubbs, A. Heine, K. Reuter, G. Klebe, "Flexible adaptations in the structure of the tRNA-modifying enzyme tRNA-guanine transglycosylase and their implications for substrate selectivity, reaction mechanism and structure-based drug design" *Chembiochem* **2003**, *4*, 1066–1077.
[16] K. Sievers, P. Neumann, L. Sušac, S. D. Vela, M. Graewert, S. Trowitzsch, D. Svergun, R. Tampé, R. Ficner, "Structural and functional insights into tRNA recognition by human tRNA guanine transglycosylase" *Structure* **2024**, *32*, 316-327.e5.
[17] J. Abramson, J. Adler, J. Dunger, R. Evans, T. Green, A. Pritzel, O. Ronneberger, L. Willmore, A. J. Ballard, J. Bambrick, S. W. Bodenstein, D. A. Evans, C.-C. Hung, M. O'Neill, D. Reiman, K. Tunyasuvunakool, Z. Wu, A. Žemgulytė, E. Arvaniti, C. Beattie, O. Bertolli, A. Bridgland, A. Cherepanov, M. Congreve, A. I. Cowen-Rivers, A. Cowie, M. Figurnov, F. B. Fuchs, H. Gladman, R. Jain, Y. A. Khan, C. M. R. Low, K. Perlin, A. Potapenko, P. Savy, S. Singh, A. Stecula, A. Thillaisundaram, C. Tong, S. Yakneen, E. D. Zhong, M. Zielinski, A. Žídek, V. Bapst, P. Kohli, M. Jaderberg, D. Hassabis, J. M. Jumper, "Accurate structure prediction of biomolecular interactions with AlphaFold 3" *Nature* **2024**, *630*, 493–500.

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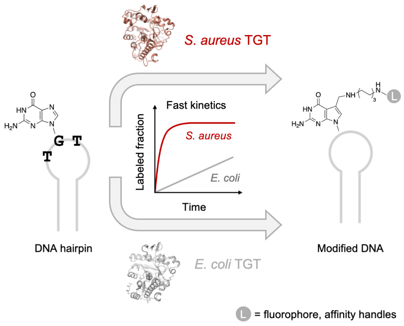

Here we present *Staphylococcus aureus* tRNA guanine transglycosylase (TGT) as a superior enzyme for site-specific covalent DNA labeling. By rationally optimizing a tRNA^{His}-derived DNA hairpin, *S. aureus* TGT achieves near-quantitative incorporation of preQ1 derivatives. We demonstrate the optimized DNA substrate enables rapid single- and multi-site labeling at any position in ssDNA constructs, greatly expanding the DNA-TAG (transglycosylation at guanosine) toolbox for imaging, diagnostics and therapeutics.